JARVIS VOICE ASSISTANT

Revised Implementation Plan — MVP-First Edition

Prepared following an architectural audit and scope-reduction review. Reduces the original 19-phase roadmap to a 10-phase MVP core loop (Phases 0–9), with remaining capabilities held in a deferred backlog.

1. Purpose & Guiding Principle

This plan supersedes the original 19-phase JARVIS roadmap. The audit found meaningful scope creep: infrastructure built for capabilities that don't exist yet (proactive alerts before there's anything to alert on), fragmented features solving the same problem twice, and complexity sequenced ahead of a proven core system.

Guiding principle: build the loop, then earn the right to add autonomy. Every deferred feature below depends on the core loop already being reliable — building them earlier is negative ROI, not acceleration.

The Core Loop

The one milestone that matters before anything else: JARVIS hears a command, understands it, selects a tool, safely executes it, receives the result, and speaks the result back — with streaming and interruption support throughout.

Example: "What's my GPU temperature?" → Deepgram (STT) → Intent Router → gpu_status() → Tool Registry → Permission Check → GPU Tool → "67°C" → Ollama → ElevenLabs → User.

2. Scope Reduction Summary

- Phases: 19 → 10 (Phase 0 through Phase 9).
- 5 consolidations: App+Media, Failover+Recovery, Fast Path+Weather/Context, Tool Registry+basic Approval, Baseline+Observability.
- 6 items moved to a deferred backlog: full Terminal/Filesystem, full Approval System, Agent Planner, Event Bus/Attention Manager/Proactive Assistant, long-term memory, Email/Calendar workflows.
- Net effect: roughly 40–50% of original scope deferred — almost entirely cross-cutting state machines and integrations with no payoff until the core loop is trustworthy.

3. Revised Phase Plan (MVP)

Ph.	Name	Scope	Note
0	Baseline & Tracing	Freeze current build. Measure STT/LLM/TTS/total latency and Ollama CPU/RAM/GPU use. Instrument full trace: USER_SPEECH → STT_FINAL → LLM_START → TOOL_CALL → TOOL_RESULT → LLM_FIRST_TOKEN → TTS_START → AUDIO_FIRST_BYTE → COMPLETE.	*Foundation — everything else is measured against this data.*
1	Architecture Refactor	Separate voice, agent, tools, memory, monitoring, and security concerns while keeping the current bot working throughout.	*Unchanged from original plan — prerequisite for scaling.*
2	Tool Registry	ToolRegistry: name, description, input schema, permission level, handler, timeout. Flow: LLM → Tool Call → Registry → Validate → Permission Check → Execute. First tools: get_time,	*Permission-level field absorbs simple approvals; full ApprovalManager deferred.*

Ph.	Name	Scope	Note
		get_system_info, get_gpu_info, get_weather, media_control.	
3	Fast Path + Context Tools	IntentRouter: DIRECT_TOOL / AGENT_TASK / CONVERSATION / UNKNOWN. Route deterministic commands (time, date, weather, system status, GPU, media, open app) straight to tools, bypassing LLM reasoning.	*Merged with original Weather/Context phase — weather was duplicated.*
4	App & Media Control	Single AppController: resolve name (alias + fuzzy match) → launch or focus → confirm running state. Media actions (play/pause/next/prev/volume/mute) via one default provider.	*Merged App Orchestrator + Media Controller — identical resolve/execute shape.*
5	On-Demand System Monitoring	CPU/GPU/RAM/VRAM/disk stats served as pull-based tool calls, not a continuous background daemon.	*Trimmed — continuous polling has no consumer pre-MVP.*
6	Resilience & Failover	HealthManager tracks LiveKit, Deepgram, ElevenLabs, Ollama, internet with HEALTHY/DEGRADED/DOWN. Bounded retries, timeouts, one deterministic fallback (e.g. local model on primary failure).	*Merged Emergency Failover + Recovery/Self-Healing.*
7	Short-Term Memory	Conversation memory plus working memory: current app, project, task, pending tool.	*Long-term preference store deferred until usage data exists.*
8	Voice Experience Optimization	Streaming STT, early intent detection, streaming LLM output, sentence-level TTS, interruption cancellation. Optimize for time-to-first-audio, not total generation time.	*Unchanged — directly serves the core-loop milestone.*
9	Observability	Aggregate Phase 0 trace data: latency percentiles (avg/p95/p99), tool success/failure rate, fallback rate, interruption count.	*Merged into Phase 0 pipeline — dashboard layer, not a new system.*

4. Recommended Sequencing

Phase 0 → Phase 1 → Phase 2 → Phase 3 → Phase 4 → Phase 5 → Phase 6 → Phase 7 → Phase 8 → Phase 9

Phases 4–5 (App/Media, Monitoring) and 6–7 (Resilience, Memory) can run in parallel workstreams once Phase 3's Fast Path is stable, since neither pair has a hard dependency on the other.

5. Deferred Backlog (Post-MVP)

Build only after the core loop above is proven reliable in daily use. Sequencing within the backlog is not fixed — prioritize by what real usage demands.

Deferred Item	Why It Waits
Full Terminal / Filesystem Suite	Search, copy, move, directory creation, PDF extraction — large safety surface; MVP only needs read-only status checks.
Full Approval System	Operation-bound confirmation workflow for destructive/external actions. Not needed while MVP tools are read-only/low-risk.

Deferred Item	Why It Waits
Agent Planner	Multi-step task graphs (Planner/Executor/TaskState). Highest complexity item in the original plan; no tool ecosystem yet to plan over.
Event Bus + Attention Manager + Proactive Assistant	Build as one combined effort once there's something worth being proactive about.
Long-Term Memory + Context Manager	Preference storage and relevance-based retrieval. Needs real usage data to design well.
Email / Calendar Workflows	Separate integration surface (OAuth, drafts, send/approve) unrelated to the core voice-loop problem.

6. Exit Criteria for MVP

- Core loop (hear → understand → select tool → execute → speak) works reliably for all Phase 2–3 tools with streaming and interruption.
- p95 total response latency and time-to-first-audio are measured and trending down, not just tracked.
- At least one full failover event (e.g. Ollama down → fallback) has been tested and recovers without manual intervention.
- No destructive or external-action tools exist yet — so the full Approval System has nothing to gate and can safely stay deferred.